Log-Rank Test

Log-Rank Test

Compare survival over time between two or more groups (nonparametric).

Null Hypothesis
The groups have the same survival probabilities over time.

Alternative Hypothesis
The groups have different survival probabilities over time.

Demo

Log-Rank Test

Knowledge Check

True or False:

Based on the log-rank test results we reject the null hypothesis at a 0.05 significance level.

```
Call:
survdiff(formula = Surv(mos_to_death, died) ~ female, data =
hrs_df)

n=2394, 23 observations deleted due to missingness.

            N Observed Expected (O-E)^2/E (O-E)^2/V
female=0  936      325      271     10.98      18.7
female=1 1458      354      408      7.27      18.7

 Chisq= 18.7  on 1 degrees of freedom, p= 2e-05
```

Knowledge Check

True or False:

Based on the log-rank test results we reject the null hypothesis at a 0.05 significance level.

```
Call:
survdiff(formula = Surv(mos_to_death, died) ~ female, data =
hrs_df)

n=2394, 23 observations deleted due to missingness.

            N Observed Expected (O-E)^2/E (O-E)^2/V
female=0  936      325      271     10.98      18.7
female=1 1458      354      408      7.27      18.7

 Chisq= 18.7  on 1 degrees of freedom, p= 2e-05
```



University of California
San Francisco